CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

BRAF / MEK: 7 molecules stopped early, all small molecules

9 stopped trials, but 7 molecules — all small molecules: Cobimetinib, Regorafenib, Trametinib, Binimetinib and 3 more. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

4.3%
9 of 210 closed trials (95% CI 2.3%–7.9%)

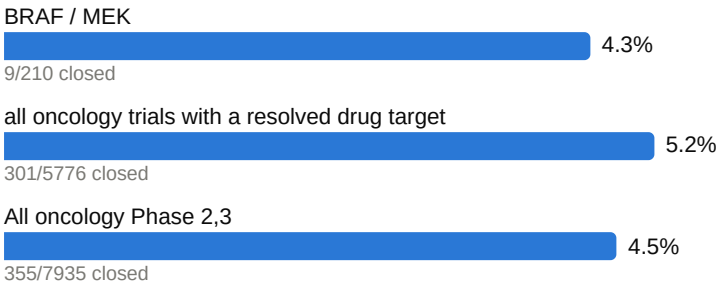
3 / 1 / 1
efficacy / safety / both (adds to 9)

43.8%
of 479 trials have closed · 269 still open or unresolved

1.7%
stopped for a biological reason within 3 years of starting (95% CI 0.8%–3.7%; 258 still at risk; 1.5% area-wide)

4 of 5 **Completed and missed the primary endpoint.** Of 146 completed trials in this segment, 5 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 4 missed and 1 met. Counted separately and never in the rate above; a floor, not a rate. 64 of the 135 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 277 distinct experimental drugs across 479 trials (small molecule 479, antibody 178, unknown 58, adc 16). Most tested: REGORAFENIB (108), TRAMETINIB (86), SORAFENIB (75), BINIMETINIB (65), COBIMETINIB (60). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 9 stops came from 9 sponsor-asset programmes across 9 sponsors; the largest (UNICANCER) contributed 1; removing that programme's trials from both sides leaves 3.8%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 11 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 9.5%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	2	93	2.2%	0.6%–7.5%
2018–2020	3	72	4.2%	1.4%–11.5%
2021–2024	4	45	8.9%	3.5%–20.7%

Where the stops sit

Benefit–risk: 4, Efficacy / futility: 3, Safety: 1, Efficacy and safety: 1. Sponsors with most stops: Cancer Trials Ireland (1), Great Novel Therapeutics Biotech & Medicals Corporation (1), SpringWorks Therapeutics, Inc., a healthcare company of Merck KGaA, Darmstadt, Germany (1), Bristol Myers Squibb (1), UNICANCER (1), Columbia University (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05518110	2	Cancer Trials Ireland	TRAMETINIB, HYDROXYCHLOROQUINE	2026-03	Efficacy	Study terminated due to futility

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05770882	2	Great Novel Therapeutics Biotech & Medicals Corporation	TUCIDINOSTAT, REGORAFENIB	2025-01	Benefit–risk	The overall profile does not support development for Hepatocellular Carcinoma
NCT05580770	2	SpringWorks Therapeutics, Inc., a healthcare company of Merck KGaA, Darmstadt, Germany	MIRDAMETINIB, BRIMARAFENIB	2025-01	Safety	Lack of tolerability of BGB-3245 (Brimarafenib) when given in combination with Mirdametinib
NCT05071183	2	Bristol Myers Squibb BMY	REPOTRECTINIB, TRAMETINIB	2023-03	Benefit–risk	Inability to achieve and optimize a dose that could provide a positive benefit risk profile
NCT03722108	2	UNICANCER	REGORAFENIB, IRINOTECAN	2022-05	Benefit–risk	Unfavorable benefit-risk balance in the experimental arm following the IDMC for intermediate efficacy and safety analysis
NCT04214418	2	Columbia University	COBIMETINIB, HYDROXYCHLOROQUINE, ATEZOLIZUMAB	2022-04	Efficacy + safety	Toxicity and Lack of Efficacy

The 6 most recent of 9 stopped trials. Every molecule behind all 9 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 43.8% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.